

Project 1D

How the Second Twist Angle Reshapes the Global Kinematics of CuRo3R

Motivation

Cuspidality is a global property of a robot, and it can change when seemingly small geometric parameters are modified. In a 3R manipulator, the twist between consecutive joint axes determines how the joint-space singularity set folds into workspace and therefore how inverse-kinematic solutions are created, merged, separated, or connected.

In this project you will use the supplied CuRo3R robot as a reference design and create a controlled one-parameter family. All DH parameters are kept equal to those obtained from the supplied CuRo3R model, except for α_2 .

The mandatory geometries are

$$\alpha_2 \in \left\{ \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2} \right\}.$$

You are encouraged to investigate additional values of α_2 whenever they help locate a transition or explain a change in the global kinematic structure.

The central question is:

How does changing only α_2 modify the singularity topology, IK distribution, cuspidality, and feasible posture-changing motions of the same 3R robot?

Task

Begin from the supplied URDF and reconstruct a consistent DH and screw representation of CuRo3R. This reference model defines the geometric parameters that remain unchanged throughout the project. You must show that your DH and Product-of-Exponentials models describe the same mechanism by comparing their forward kinematics with the URDF model.

Next, formulate the kinematics as a family parameterized by α_2 . Wherever possible, keep α_2 symbolic long enough to reveal how it enters the forward kinematics, the screw-based positioning Jacobian 0J_3 , and the singularity condition. The four mandatory values above must then be studied using exactly the same conventions and numerical procedures so that the resulting global maps can be compared directly.

For each value of α_2 , derive and implement the complete analytical inverse kinematics. The solver must return every real configuration associated with a target position, correctly account for angular periodicity, and verify every candidate through forward kinematics. Use the same representative workspace targets across the parameter sweep whenever possible; this will make changes in IK multiplicity and branch organization visible rather than hiding them behind different test cases.

The main analysis is global. For every mandatory α_2 , construct the singularity curves in the (q_2, q_3) plane and map them into workspace. Use these maps to determine the regions with different numbers of inverse solutions and to study how those regions evolve as α_2 changes. Select representative targets in the important regions and place all of their IK solutions back onto the joint-space maps.

The final part of the project is to relate these geometric changes to cuspidality and motion planning. For each geometry, investigate whether selected distinct IK solutions can be connected while remaining inside the regular configuration space. Use comparable start/goal configurations or workspace paths across the parameter sweep. If a nonsingular posture change exists, construct and verify it. If it does not, identify the singularity barrier that prevents it. Additional values of α_2 may be used to locate more precisely where the observed global behavior changes.

Expected Output

- A validated reference DH and screw model reconstructed from the supplied CuRo3R URDF.
- A clearly defined parameterized robot family with $a_1 = 1$, all other DH parameters fixed to the CuRo3R values, and α_2 varied.
- Numerical cross-validation of URDF, DH, and PoE forward kinematics for the reference model and validation of the modified models.
- A screw-based derivation and numerical verification of 0J_3 , showing explicitly how α_2 affects the Jacobian and singularity condition.
- Complete analytical inverse kinematics for each mandatory value $\alpha_2 \in \{\pi/6, \pi/4, \pi/3, \pi/2\}$.
- Comparable (q_2, q_3) singularity maps for all mandatory values of α_2 .
- Corresponding workspace critical curves and regions with different numbers of inverse-kinematic solutions.
- Representative targets with all corresponding IK solutions located on the joint-space maps.
- A parameter-comparison figure or atlas showing how the singularity topology and IK distribution evolve with α_2 .
- A connectivity-based classification of the observed cuspidal or noncuspidal behavior for each mandatory geometry.
- Branch-preserving and posture-change motion experiments that relate path feasibility to the global maps.
- Additional α_2 cases, where useful, to refine or explain transitions observed between the mandatory geometries.

Useful resources

1. Simulations to visualize your robots: <https://epfl-lasa.github.io/eng-654/#simulations>
2. Robot assets (URDF and .stl files for visualization): <https://epfl-lasa.github.io/eng-654/#download-assets>